0 vimrc

vimrc

```
"variable
let mapleader = ','

"
set number
set relativenumber
set noexpandtab
set shiftwidth=4
set tabstop=4

inoremap kj <Esc>
inoremap {<CR> {}<Left><CR><Esc>O
tnoremap kj <C-\><C-n>
nnoremap <leader>c :w<CR>:!g++ -std=c++20 % -o %:r && ./%:r<CR>
nnoremap ' `
nnoremap ` '
```

1 Number Theory

Exgcd.cpp

```cpp
array<ll,2> exgcd(ll a, ll b){
        if(b == 0) return array<ll,2>{1,0};
        auto ans = exgcd(b, a % b);
        return array<ll,2>{ans[1], ans[0] - a / b * ans[1]};
}

```

catalan.cpp

```
C_0=1
C_n = $\sum C_i * C_{n-1-i}$ = $\frac{1}{n+1} C_n^{2*n}$
```

findPrime.cpp

```cpp
constexpr ll mxN=1e6+1;
mobius[1]=etf[1]=1;
forn(i,2,mxN){
        if(!prime[i]) prime[i]=i,mobius[i]=-1,etf[i]=i-1,st[++sz]=i;
        forn(j,1,sz+1){
                if(i*st[j]>=mxN) break;
                prime[i*st[j]]=st[j];
                if(i%st[j]==0){
                        mobius[i*st[j]]=0;
                        etf[i*st[j]]=etf[i]*st[j];
                        break;
                }
                else mobius[i*st[j]]=mobius[i]*-1,etf[i*st[j]]=etf[i]*(st[j]-1);
        }
}
```

matrixMultiple.cpp

```cpp
// mod 需外部定義; 快速冪用時記得單位矩陣
vector<vector<ll>> matrixMultiply(const vector<vector<ll>>& A, const vector<vector<ll>>& B) {
    ll rowA = A.size();
    ll colA = A[0].size();
    ll colB = B[0].size();
    vector<vector<ll>> C(rowA, vector<ll>(colB, 0));
    for (ll i = 0; i < rowA; ++i) {
        for (ll k = 0; k < colA; ++k) {
            for (ll j = 0; j < colB; ++j) {
                C[i][j] = (C[i][j] + A[i][k] * B[k][j]) % mod;
            }
        }
    }
    return C;
}
```

ntt.cpp

```cpp
using ll=long long;
// 注意: 兩模數CRT, 結果是真值 mod 998244353*1004535809 (~1.0028e18)
// 真實係數必須 < 該值 (值域1e9且長度>~1000就會爆), 否則需三模數NTT
// 長度上限 2^21 (n+m <= ~2e6): fft2 模數只有 2^21 階單位根, 超過會靜默出錯
 class FFT{
        public:
                ll mod,root,root_inv,maX,M,M_inv;
                FFT(){}
                FFT(ll a,ll b,ll c,ll d,ll e,ll f):mod(a),root(b),root_inv(c),maX(d),M(e),M_inv(f){}
                ll fsp(ll a,ll m){
                        ll ans=1;
                        while(m){
                                if(m&1) ans=(ans*a)%mod;
                                a=(a*a)%mod,m>>=1;
                        }
                        return ans;
                }
                void fft(ll sz,vc<ll> &arr,ll inv){
                        for(ll i=1,j=0;i<sz;++i){
                                ll bit=sz>>1;
                                for(;j&bit;bit>>=1) j^=bit; j^=bit;
                                if(i<j) swap(arr[i],arr[j]);
                        }
                        for(ll l=2;l<=sz;l<<=1){
                                ll wlen=inv?root_inv:root;
                                for(ll i=l;i<maX;i<<=1) wlen=(wlen*wlen)%mod;
                                for(ll i=0;i<sz;i+=l){
                                        ll w=1;
                                        for(ll j=0;j<l/2;++j){
                                                ll x=arr[i+j],y=(w*arr[i+j+l/2])%mod;
                                                arr[i+j]=(x+y<mod?x+y:x+y-mod);
                                                arr[i+j+l/2]=(x-y>=0?x-y:x-y+mod);
                                                w=(w*wlen)%mod;
                                        }
                                }
                        }
                        if(inv){
                                ll n_1=fsp(sz,mod-2);
```

```cpp
                                forn(i,0,sz){
                                        arr[i]=(arr[i]*n_1)%mod;
                                }
                        }
                }
};
constexpr ll MM=1002772198720536577;
ll mul(ll a,ll b){
        __int128 aa=a,bb=b,cc;
        cc=(aa*bb)%MM;
        return (ll)cc;
}
void solve(){
        FFT fft1(998244353,15311432,469870224,1<<23,1004535809,332747959);
        FFT fft2(1004535809,702606812,700146880,1<<21,998244353,669690699);
        ll n,m;
        cin >> n >> m;
        vc<ll> a(n+1),b(m+1);
        for(auto &v:a) cin >> v;
        for(auto &v:b) cin >> v;
        ll sz=1;
        for(;sz<a.size()+b.size();sz<<=1);
        a.resize(sz),b.resize(sz);
        vc<ll> A=a,B=b;
        fft1.fft(sz,a,0);
        fft1.fft(sz,b,0);
        forn(i,0,sz) a[i]=(a[i]*b[i])%fft1.mod;
        fft1.fft(sz,a,1);

        fft2.fft(sz,A,0);
        fft2.fft(sz,B,0);
        forn(i,0,sz) A[i]=(A[i]*B[i])%fft2.mod;
        fft2.fft(sz,A,1);

        forn(i,0,n+m+1) cout << (mul(a[i],fft1.M*fft1.M_inv) + mul(A[i],fft2.M*fft2.M_inv))%MM << " \n"[i==n+m];
}
int main()
{
    fast_io;
    int testcase;
    // cin>>testcase;
    testcase=1;
    while(testcase--)
        solve();
    return 0;
}
```

system-of-linear-eqn.cpp

```cpp
// mod mdl1 下高斯消去. 無解輸出-1; 無窮多解「不會偵測」(自由變數=0給特解)
// 要偵測無窮多解: 消去結束後 pivot 數 l < m 即有自由變數 => 無窮多解
// 輸入係數需已在 [0,mdl1) 內 (負數要先轉正); inv 其實是快速冪, 模數寫死 mdl1
constexpr ll mxN=505;
ll n,m;
ll inv(ll a,ll m){ll ans; for(ans=1;m;ans=(m&1?(ans*a)%mdl1:ans),a=(a*a)%mdl1,m>>=1); return ans;}
void solve(istream &cin){
        cin>>n>>m;
        vc<ll> res(m,0);
        vc<vc<ll>> a(n,vc<ll>(m+1));
        forn(i,0,n) forn(j,0,m+1) cin>>a[i][j];
        ll l=0;
        for(ll i=0;i<m && l<n;++i){
                if(!a[l][i]) forn(j,l+1,n) if(a[j][i]){swap(a[j],a[l]); break;}
                if(!a[l][i]) continue;
                ll c=inv(a[l][i],mdl1-2);
                for(auto&v:a[l]) v=(v*c)%mdl1;
                forn(j,l+1,n){
                        ll co=a[j][i];
                        forn(k,i,m+1) a[j][k]=((a[j][k]-co*a[l][k])%mdl1+mdl1)%mdl1;
                }
                l++;
        }
        forr(i,n-1,0){
                ll j=0;
                while(j<m+1 && !a[i][j]) j++;
                if(j==a[i].size()) continue;
                if(j+1==a[i].size()){cout << -1 << '\n'; return;}
                ll u=a[i][m];
                forn(k,j+1,m) u=((u-a[i][k]*res[k])%mdl1+mdl1)%mdl1;
                res[j]=u;
        }
        forn(i,0,m) cout << res[i] << " \n"[i==m-1];
}
int main()
{
    fast_io;
    ll testcase;
    // cin>>testcase;
    testcase=1;
    while(testcase--)
        solve(cin);
    return 0;
}
```

2 Tree

LCA.cpp

```cpp
constexpr ll mxN=2e5+1;
ll pa[mxN][20],d[mxN]{};
vc<ll> adj[mxN];
void dfs(ll u=1,ll p=0){d[u]=d[p]+1,pa[u][0]=p; for(auto&v:adj[u]) if(v!=p) dfs(v,u);}
void init(ll n){ // 建好 adj 後呼叫一次
        dfs();
        forn(j,1,20) forn(i,1,n+1) pa[i][j]=pa[pa[i][j-1]][j-1];
}
ll lca(ll u,ll v){
        if(d[u]<d[v]) swap(u,v);
        forr(i,19,0) if(d[u]-(1<<i)>=d[v]) u=pa[u][i];
        if(u==v) return u;
        forr(i,19,0) if(pa[u][i]!=pa[v][i]) u=pa[u][i],v=pa[v][i];
        return pa[u][0];
}
```

MST.cpp

```cpp
//kruskal
//vertex is 1 index
#define mls multiset

ll n,pa[mxN];//memset(pa,-1,sizeof(pa));
mls<ar<ll,3>> adj;

ll find(ll x){return pa[x]<0?x:pa[x]=find(pa[x]);}
void un(ll x,ll y){
    x=find(x),y=find(y); if(x==y) return;
    if(-pa[x]>-pa[y]) swap(x,y); pa[y]+=pa[x],pa[x]=y;
}
ll kruskal(){ // 回傳 MST 總權重; 結束後 cnt<n-1 代表圖不連通
    ll cnt=0,wi=0;
```

```cpp
    while(!adj.empty() && cnt<n-1){
        auto [w,u,v]=*adj.begin(); adj.erase(adj.begin());
        if(find(u)!=find(v)) wi+=w,un(u,v),cnt++;
    }
    return wi;
}
```

binary-lifing.cpp

```cpp
constexpr ll mxN=2e5+1;
ll n,q,p[mxN][20]={0}; //深度<mxN<2^18，20層夠(49MB->32MB); n更大要加層數
vc<ll> adj[mxN];
void solve(istream &cin){
    cin>>n>>q;
    forn(i,2,n+1) cin>>p[i][0];
    forn(j,1,20) forn(i,1,n+1) p[i][j]=p[p[i][j-1]][j-1];
    while(q--){
        ll x,k; cin>>x>>k;
        k=min(k,(ll)mxN); // 跳超過深度必到0, clamp避免高位bit被忽略
        forn(i,0,20) if(k>>i&1) x=p[x][i];
        cout << (x?x:-1) << '\n';
    }
}
```

find-centriod.cpp

```cpp
void dfs(ll u=1,ll p=0){
    d[u]=1;
    for(auto&v:adj[u]){
        if(v==p)
            continue;
        dfs(v,u);
        d[u]+=d[v];
    }
}
ll dfs_solve(ll u=1,ll p=0){
    for(auto&v:adj[u]){
        if(v==p)
            continue;
        if(d[v]*2>n) return dfs_solve(v,u);
    }
    return u;
}
```

hld.cpp

```cpp
constexpr ll mxN=2e5+1;
ll tin=0,in[mxN]{},t[mxN<<1];
ll n,q,a[mxN],d[mxN]{},e[mxN]{};
vc<ll> adj[mxN];
ll dfs(ll u=1,ll p=0){
        ll mx=0,sub=1;
        forn(i,0,adj[u].size()){
                ll v=adj[u][i];
                if(v!=p){d[v]=d[u]+1; ll val=dfs(v,u); sub+=val; if(val>mx) mx=val,e[u]=v;}
                else swap(adj[u][i],adj[u][0]);
        }
        return sub;
}
void hld(ll u=1,ll p=1){
        in[u]=++tin,t[in[u]+n-1]=a[u];
        if(e[u]) hld(e[u],p);
        forn(i,1,adj[u].size()){ll v=adj[u][i]; if(e[u]!=v) hld(v,v);}
        e[u]=p;
}
void pst(ll p,ll val){for(t[p+=n-1]=val;p>>=1;t[p]=max(t[p<<1],t[p<<1|1]));}

ll qry(ll l,ll r){
        ll ans=0;
        for(l+=n-1,r+=n-1;l<=r;l>>=1,r>>=1){if(l&1) ans=max(ans,t[l++]); if(r&1^1) ans=max(ans,t[r--]);}
        return ans;
}
ll query(ll a,ll b){
        ll ans=0;
        for(;e[a]!=e[b];b=adj[e[b]][0]){
                if(d[e[a]]>d[e[b]]) swap(a,b);
                ans=max(ans,qry(in[e[b]],in[b]));
        }
        if(in[a]>in[b]) swap(a,b);
        return max(ans,qry(in[a],in[b]));
}
void solve(istream &cin){
        cin >> n >> q;
        forn(i,1,n+1) cin >> a[i];
        forn(i,0,n-1){ll u,v; cin >> u >> v; adj[u].emp(v),adj[v].emp(u);}
        adj[1].emp(0);
        dfs();
        hld();
        forr(i,n-1,1) t[i]=max(t[i<<1],t[i<<1|1]);
        for(ll op,a,b;q--;){
                cin >> op >> a >> b;
                if(op==1) pst(in[a],b);
                else cout << query(a,b) << ' ';
        }
}
```

small-to-large.cpp

```cpp
// 注意：每層遞迴持有deque，鏈狀樹會爆棧且記憶體~300MB，深樹題慎用
constexpr ll mxN=2e5+1;
ll n,k,res=0;
vc<ll> adj[mxN];
deque<ll> dfs(ll u=1,ll p=0){
        deque<ll> cur(1,{1}),tmp;
        for(auto &v:adj[u]){
                if(v!=p){
                        tmp=dfs(v,u),tmp.push_front(0);
                        if(tmp.size()>cur.size()) swap(tmp,cur);
                        forn(i,0,tmp.size()) if(k-i>=0 && k-i<cur.size()) res+=cur[k-i]*tmp[i];
                        forn(i,0,tmp.size()) cur[i]+=tmp[i];
                }
        }
        return cur;
}
void solve(istream &cin){
        cin >> n >> k;
        forn(i,0,n-1){ll u,v; cin >> u >> v; adj[u].emp(v),adj[v].emp(u);}
        dfs();
        cout << res << '\n';
}
```

tree-diameter.cpp

```cpp
#define emp emplace_back
constexpr ll mxN=2e5+1;
ll n,res=0;
vc<ll> adj[mxN],dp(mxN);

void dfs(ll u=1,ll p=0){dp[u]=0; for(auto&v:adj[u]) if(v!=p) dfs(v,u),res=max(res,dp[u]+dp[v]+1),dp[u]=max(dp[u],dp[v]+1);}
```

```
int main()
{
    cin>>n; forn(i,0,n-1){ll u,v; cin>>u>>v; adj[u].emp(v),adj[v].emp(u);}
    dfs();
    cout << res << '\n';
}
```

3 Graph

2-sat.cpp

```
// 遞迴深度可達 2m(=2e5), 注意 stack 限制
constexpr ll mxN=2e5+1;
ll sc=0,scc[mxN];
ll id=0,st[mxN],vs[mxN]{};
ll n,m,tin=0,dfn[mxN]{},low[mxN];
vc<ll> adj[mxN];
void tarjan(ll u){
	dfn[u]=low[u]=++tin,st[++id]=u,vs[u]=1;
	for(auto &v:adj[u]){
		if(!dfn[v]) tarjan(v),low[u]=min(low[u],low[v]);
		else if(vs[v]) low[u]=min(low[u],dfn[v]);
	}
	if(low[u]==dfn[u]){
		sc++;
		for(;id;){
			scc[st[id]]=sc,vs[st[id]]=0;
			id--;
			if(st[id+1]==u) break;
		}
	}
}
void solve(istream &cin){
	cin >> n >> m;
	forn(i,1,n+1){
		char c1,c2; ll x,y,X,Y; cin >> c1 >> x >> c2 >> y;
		X=x+m,Y=y+m;
		if(c1=='-') swap(x,X);
		if(c2=='-') swap(y,Y);
		adj[X].emp(y),adj[Y].emp(x);
	}
	forn(i,1,2*m+1) if(!dfn[i]) tarjan(i);
	forn(i,1,m+1) if(scc[i]==scc[i+m]){cout << "IMPOSSIBLE" << '\n'; return;}
	forn(i,1,m+1) cout << (scc[i]>scc[i+m]?'-':'+') << " \n"[i==m];
}
```

BellmanFord.cpp

```
//SPFA; 回傳false=有負環(僅偵測從start可達的); d[]需外部初始化INF, d[start]=0
bool spfa(){
    vc<ll> vs(n+1,0); //1-indexed 要開 n+1
    queue<ll> q; q.emplace(start);
    while(!q.empty()){
        ll u=q.front(); q.pop(),vs[u]=0;
        for(auto&[v,w]:adj[u]){
            if(d[v]>d[u]+w){
                d[v]=d[u]+w;
                if(!vs[v]){vs[v]=1,cnt[v]++; if(cnt[v]>n) return false; q.emplace(v);}
            }
        }
    }
    return true;
}
```

Dijkstra.cpp

```
// 前置: res[]全設INF、起點寫死1(res[1]=0); lazy-deletion寫法
void djk(){
	mls<pll> hp;
	res[1]=0,hp.emplace(pll{res[1],1});
	while(!hp.empty()){
		auto [d,u]=*hp.bg(); hp.erase(hp.bg());
		if(d>res[u]) continue;
		for(auto&[v,w]:adj[u]){
			if(w+res[u]<res[v]) res[v]=w+res[u],hp.emplace(pll{res[v],v});
		}
	}
}
```

Edmond.cpp

```
constexpr ll mxN=505;
ll n,m,adj[mxN][mxN]{},g[mxN][mxN]{};//adj=capacity
vc<ll> pa;
ll bfs(){
	queue<pll> q; q.emplace(pll{0,inf});
	while(!q.empty()){
		auto [u,flow]=q.front(); q.pop();
		forn(i,0,n)
			if(g[u][i] && adj[u][i] && pa[i]==-1){
				pa[i]=u;
				if(i==n-1) return min(flow,adj[u][i]);
				q.emplace(pll{i,min(flow,adj[u][i])});
			}
	}
	return 0;
}
void solve(istream &cin){
	cin>>n>>m;
	pa.resize(n);
	forn(i,1,m+1){ll u,v; cin>>u>>v,u--,v--; adj[u][v]++,g[u][v]++,adj[v][u]++,g[v][u]++;} //重邊容量/重數要累加
	ll f=0;
	fill(all(pa),-1),pa[0]=0;
	for(ll nf;nf=bfs();f+=nf){
		ll cur=n-1;
		while(cur) adj[pa[cur]][cur]-=nf,adj[cur][pa[cur]]+=nf,cur=pa[cur];
		fill(all(pa),-1),pa[0]=0;
	}
	vc<pll> res;
	forn(i,0,n) forn(j,0,n) if(g[i][j] && pa[i]!=-1 && pa[j]==-1) forn(_,0,g[i][j]) res.emp(pll{i,j}); //重邊每條都要列
	cout << res.size() << '\n';
	for(auto&[u,v]:res) cout << u+1 << ' ' << v+1 << '\n';
}
int main()
{
    fast_io;
    ll testcase;
    // cin>>testcase;
    testcase=1;
    while(testcase--)
        solve(cin);
    return 0;
}
```

articulation-point.cpp

```cpp
//undirected graph articulation point; root呼叫 tarjan(1,0), 不連通圖每個分量都要跑
ll n,tim=0,art=0;
vc<ll> dfn(mxN,0),low(mxN);
void tarjan(ll u,ll p){
    ll cnt=0; bool isArt=false;
    dfn[u]=low[u]=++tim;
    for(auto&v:adj[u]){
        if(!dfn[v]){
            tarjan(v,u);
            low[u]=min(low[u],low[v]);
            if(low[v]>=dfn[u] && p) isArt=true; //非root: 某子樹回不到u之上
            cnt++;
        }
        else if(v!=p) low[u]=min(low[u],dfn[v]);
    }
    if(!p && cnt>1) isArt=true; //root: 至少兩個子樹
    art+=isArt;
}
```

bridge.cpp

```cpp
constexpr ll mxN=1e5+1;
ll n,m,dfn[mxN]{},low[mxN],tin=0;
vc<ll> adj[mxN];
vc<ar<ll,2>> res;
void tarjan(ll u=1,ll p=0){
        dfn[u]=low[u]=++tin;
        bool skip=0; //父邊只跳過一次, 重邊(平行邊)才不會被誤判成橋
        for(auto &v:adj[u]){
                if(v==p && !skip){skip=1; continue;}
                if(dfn[v]) low[u]=min(low[u],dfn[v]);
                else{
                        tarjan(v,u),low[u]=min(low[u],low[v]);
                        if(low[v]>dfn[u]) res.emp(ar<ll,2>{u,v});
                }
        }
}
```

dinic.cpp

```cpp
// 多測時 edg/adj/sz 是全域不會自動重置, 要自己清
constexpr ll mxN=505;
class fe{
        public:
                ll u,v,cp,fw;
                fe(){}
                fe(ll U,ll V,ll CP):u(U),v(V),cp(CP){fw=0;}
};
ll n,m,sz=0;
vc<fe> edg;
queue<ll> q;
vc<ll> adj[mxN],ptr,dep;
bool bfs(){
        while(!q.empty()){
                ll u=q.front(); q.pop();
                for(auto&id:adj[u]){
                        if(edg[id].cp==edg[id].fw || dep[edg[id].v]!=-1) continue;
                        dep[edg[id].v]=dep[u]+1,q.emplace(edg[id].v);
                }
        }
        return dep[n-1]!=-1;
}
ll dinic(ll u,ll push){
        if(!push) return 0;
        if(u==n-1) return push;
        for(ll &cid=ptr[u];cid<adj[u].size();++cid){
                ll id=adj[u][cid];
                ll v=edg[id].v;
                if(dep[v]!=dep[u]+1) continue;
                ll tr=dinic(v,min(push,edg[id].cp-edg[id].fw));
                if(!tr) continue;
                edg[id].fw+=tr,edg[id^1].fw-=tr;
                return tr;
        }
        return 0;
}
void solve(istream &cin){
        cin>>n>>m;
        ptr.resize(n),dep.resize(n);
        forn(i,1,m+1){
                ll u,v,c; cin>>u>>v>>c,u--,v--;
                edg.emp(fe{u,v,c}),edg.emp(fe{v,u,0});
                adj[u].emp(sz),adj[v].emp(sz+1),sz+=2;
        }
        ll res=0;
        for(;;){
                ll flow;
                fill(all(dep),-1);
                q.emplace(0),dep[0]=0;
                if(!bfs()) break;
                fill(all(ptr),0);
                for(;flow=dinic(0,inf);res+=flow);
        }
        cout << res << '\n';
}
int main()
{
    fast_io;
    ll testcase;
    // cin>>testcase;
    testcase=1;
    while(testcase--)
        solve(cin);
    return 0;
}
```

dominator-tree.cpp

```cpp
constexpr ll mxN=1e5+1;
ll n,m,tim=0,id[mxN],sd[mxN],pa[mxN],f[mxN],lb[mxN],et[mxN]{},ret[mxN];
vc<ll> adj[mxN],radj[mxN],t[mxN],st;
void dfs(ll u=1){
        et[u]=++tim,ret[tim]=u;
        id[tim]=sd[tim]=lb[tim]=pa[tim]=tim;
        for(auto&v:adj[u]){
                if(!et[v]) dfs(v),f[et[v]]=et[u];
                radj[et[v]].emp(et[u]);
        }
}
ll find(ll u,ll x=0){
        if(u==pa[u]) return x?-1:u;
        ll v=find(pa[u],x+1);
        if(v<0) return u;
        if(sd[lb[u]]>sd[lb[pa[u]]]) lb[u]=lb[pa[u]];
        pa[u]=v;
        return x?v:lb[u];
}
void built(){
        vc<ll> arr[n+1];
        forr(i,n,1){
                for(auto&v:radj[i]) sd[i]=min(sd[i],sd[find(v)]);
                if(i>1) arr[sd[i]].emp(i);
                for(auto&v:arr[i]){
                        ll x=find(v);
                        if(sd[x]==sd[v]) id[v]=sd[v];
                        else id[v]=x;
                }
```

```cpp
                if(i>1) pa[i]=f[i];
        }
        forn(i,2,n+1){
                if(id[i]!=sd[i]) id[i]=id[id[i]];
                t[ret[i]].emp(ret[id[i]]),t[ret[id[i]]].emp(ret[i]);
        }
}
void dfs_ans(ll u=1,ll p=0){
        st.emp(u);
        if(u==n){
                sort(all(st));
                cout << st.size() << '\n';
                for(auto&v:st) cout << v << ' '; cout << '\n';
                exit(0);
        }
        for(auto&v:t[u]){
                if(v==p) continue;
                dfs_ans(v,u);
        }
        st.pop_back();
}
void solve(istream &cin){
        cin>>n>>m;
        forn(i,1,m+1){
                ll u,v; cin>>u>>v;
                adj[u].emp(v);
        }
        dfs();
        built();
        dfs_ans();
}
```

floyd-warshall.cpp

```cpp
// 先設 f[x][x]=0 其餘INF (INF用0x3f3f3f3f3f3f3f3f可安全相加)
for (ll k = 1; k <= n; k++) {
  for (ll x = 1; x <= n; x++) {
    for (ll y = 1; y <= n; y++) {
      f[x][y] = min(f[x][y], f[x][k] + f[k][y]);
    }
  }
}
```

hierholzer.cpp

```cpp
//debrujin sequence 特化版(n<=15), 不是通用歐拉迴路模板, 一般題要整段改寫
ll n;
string s;
vc<ll> adj[1<<14|1];
void dfs(ll u=0){
    while(!adj[u].empty()){
        ll v; v=adj[u].back(); adj[u].pop_back();
        dfs(v);
        s+=(v&1)?'1':'0';
    }
}
void solve(istream &cin){
    cin>>n;
    if(n==1){cout << "01" << '\n'; return;}
    forn(i,0,1<<(n-1)){
        adj[i].emp((i<<1)&((1<<(n-1))-1));
        adj[i].emp((i<<1|1)&((1<<(n-1))-1));
    }
    dfs();
    forn(i,0,n-1) s+='0';
    reverse(all(s));
    cout << s << '\n';
}
```

hopcroft-karp.cpp

```cpp
// mxN=501 => n,m<=500; 匹配數 0 時不會輸出任何匹配行
constexpr ll mxN=501;
ll n,m,k,mat[mxN<<1]{},d[mxN<<1];
vc<ll> adj[mxN];
ll bfs(){
        queue<ll> q;
        memset(d,-1,sizeof(d));
        forn(i,1,n+1) if(!mat[i]) d[i]=0,q.emplace(i);
        while(!q.empty()){
                ll u=q.front(); q.pop();
                if(~d[0]) return 1;
                for(auto &v:adj[u]){
                        if(!(~d[mat[v]])) d[mat[v]]=d[u]+1,q.emplace(mat[v]);
                }
        }
        return 0;
}
bool dfs(ll u){
        if(!u) return true;
        for(auto &v:adj[u]) if(d[mat[v]]==d[u]+1 && dfs(mat[v])){mat[u]=v,mat[v]=u;return true;}
        d[u]=-1;
        return false;
}
void solve(istream &cin){
        cin >> n >> m >> k;
        forn(i,0,k){ll u,v; cin >> u >> v; adj[u].emp(n+v);}
        ll res=0;
        for(;bfs();) forn(i,1,n+1) if(!mat[i]) res+=dfs(i);
        cout << res << '\n';
        if(res) forn(i,1,n+1) if(mat[i]) cout << i << ' ' << (mat[i]-n) << '\n';
}
```

kuhn.cpp

```cpp
// base=500寫死 => 隱含 n,m<=500 (女生編號=base+j), 換題記得改
constexpr ll mxN=1001;
constexpr ll base=500;
ll n,m,k,vs1[mxN],vs2[mxN],ok[mxN][mxN]{};
vc<ll> adj[mxN];
bool dfs(ll u){
        if(vs2[u]) return false;
        vs2[u]=1;
        for(auto&v:adj[u]) if(!vs1[v] || dfs(vs1[v])){vs1[v]=u; return true;}
        return false;
}
void solve(istream &cin){
        cin >> n >> m >> k;
        //[1,500] boy [500+1,500+500] girl
        forn(i,0,k){
                ll u,v; cin >> u >> v,v+=base;
                if(!ok[u][v])
                        adj[u].emp(v),adj[v].emp(u),ok[u][v]=1;
        }
        memset(vs1,0,sizeof(vs1));
        forn(i,1,n+m+1){
                memset(vs2,0,sizeof(vs2));
                dfs(i);
        }
        vc<ar<ll,2>> res;
        forn(i,1,n+1){
                for(auto &v:adj[i]){
                        if(vs1[v]==i) res.emp(ar<ll,2>{i,v-base});
                }
```

```cpp
        }
        cout << res.size() << '\n';
        for(auto&[u,v]:res) cout << u << ' ' << v << '\n';
}
```

mcmf.cpp

```cpp
// SPFA版MCMF: 邊權必須非負(無負環偵測, 原圖有負權會死迴圈); 源1匯n寫死; 多測清 e/id/adj
class Edge{
        public:
                ll u,v,c,w;
                Edge(){}
                Edge(ll u,ll v,ll c,ll w):u(u),v(v),c(c),w(w){}
};
constexpr ll mxN=501;
ll n,m,k,id=0,vs[mxN]{},d[mxN],pa[mxN];
vc<Edge> e;
vc<ll> adj[mxN];
ll bfs(){
        memset(pa,-1,8*(n+1)),memset(d,0x3f,sizeof(d));
        queue<ll> q;
        d[1]=0,vs[1]=1,q.emplace(1);
        while(!q.empty()){
                auto u=q.front(); q.pop(),vs[u]=0;
                for(auto &eid:adj[u]){
                        auto [x,v,c,w]=e[eid];
                        if(c && d[u]+w<d[v]){
                                pa[v]=eid,d[v]=d[u]+w;
                                if(!vs[v])
                                        vs[v]=1,q.emplace(v);
                        }
                }
        }
        return d[n]<INF;
}
void solve(){
        cin >> n >> m >> k;
        forn(i,1,m+1){
                ll u,v,c,w; cin >> u >> v >> c >> w;
                e.emp(Edge{u,v,c,w}),e.emp(Edge{v,u,0,-w}),adj[u].emp(id++),adj[v].emp(id++);
        }
        ll f=0,res=0;
        for(ll nf,v;f<k && bfs();){
                for(nf=k-f,v=n;v!=1;nf=min(nf,e[pa[v]].c),v=e[pa[v]].u);
                for(v=n;v!=1;e[pa[v]].c-=nf,e[pa[v]^1].c+=nf,v=e[pa[v]].u);
                res+=d[n]*nf,f+=nf;
        }
        cout << (f<k?-1:res) << '\n';
}
```

scc.cpp

```cpp
// scc編號從0開始(2-sat.cpp裡的版本是先++再給, 兩者編號差1, 混用注意)
void dfs(ll u){
        dfn[u]=low[u]=++tim,st[++id]=u,vs[u]=1;
        for(auto &v:adj[u]){
                if(!dfn[v])
                        dfs(v),low[u]=min(low[u],low[v]);
                else if(vs[v]) low[u]=min(low[u],dfn[v]);
        }
        if(low[u]==dfn[u]){
                for(ll v=-1;v!=u;scc[st[id]]=sc,vs[st[id]]=0,v=st[id],id--);
                sc++;
        }
}
```

toposort.cpp

```cpp
void topo_sort() {
        int n, m;
        cin >> n >> m;
        vector<vector<int>> graph(n+1);
        for (int i = 0; i < m; i++) {
                int a, b;
                cin >> a >> b;
                graph[a].push_back(b);
        }
        vector<int> rudu(n+1, 0);
        for (auto &node : graph) {
                for (auto &now : node) {
                        rudu[now]++;
                }
        }
        queue<int> que;
        for (int i = 1; i <= n; i++) {
                if (rudu[i] == 0) que.push(i);
        }
        vector<int> topo;
        while (!que.empty()) {
                int top = que.front();
                que.pop();
                topo.push_back(top);
                for (auto &i : graph[top]) {
                        rudu[i]--;
                        if (rudu[i] == 0) que.push(i);
                }
        }
        if (topo.size() == n) {
                for (auto &i : topo) cout << i << " ";
        }
        else
                cout << "IMPOSSIBLE";
}
```

4 Geometry

Determine whether a polygon is convex.cpp

```cpp
// 注意: 自交但每個轉向同號的多邊形(如五角星一筆畫)會被誤判convex,
// 嚴格判定需另檢查總轉角恰為±360度; 整數座標>~3e7時double cross有精度風險, 應改ll
#include <vector>
using namespace std;

struct Point {
    double x, y;
};

double cross(const Point& a, const Point& b, const Point& c) {
    // 計算向量 ab x bc
    double x1 = b.x - a.x, y1 = b.y - a.y;
    double x2 = c.x - b.x, y2 = c.y - b.y;
    return x1 * y2 - y1 * x2;
}

bool isConvex(const vector<Point>& poly) {
    int n = poly.size();
    if (n < 3) return false;
    int sign = 0;
    for (int i = 0; i < n; ++i) {
        double c = cross(poly[i], poly[(i + 1) % n], poly[(i + 2) % n]);
        if (c != 0) {
            if (sign == 0)
                sign = (c > 0) ? 1 : -1;
            else if ((c > 0 && sign < 0) || (c < 0 && sign > 0))
```

```cpp
                return false;
            }
        }
        return true;
}
```

Rotating Calipers 求凸包直徑.cpp

```cpp
struct Point {
    double x, y;
};

double dist2(const Point& a, const Point& b) {
    double dx = a.x - b.x, dy = a.y - b.y;
    return dx * dx + dy * dy;
}

// 假設 convex 是逆時針的「嚴格」凸包(無共線點/重複點, 否則two-pointer不保證正確)
// 座標>1e7時 double 平方和超過2^53有精度風險, 應改ll算dist2最後才sqrt
double convexHullDiameter(const vector<Point>& convex) {
    int n = convex.size();
    if (n == 1) return 0;
    if (n == 2) return sqrt(dist2(convex[0], convex[1]));
    double res = 0;
    for (int i = 0, j = 1; i < n; ++i) {
        while (dist2(convex[i], convex[(j + 1) % n]) > dist2(convex[i], convex[j]))
            j = (j + 1) % n;
        res = max(res, dist2(convex[i], convex[j]));
    }
    return sqrt(res);
}
```

all solutions for Diophantine Equation in given range.cpp

```cpp
/*
    1. General solution to the linear Diophantine equation:
        a * x + b * y = c
        If (x0, y0) is a particular solution and g = gcd(a, b):
            x = x0 + k * (b / g)
            y = y0 - k * (a / g)
            for any integer k
    2. Number of integer solutions (x, y) within bounds:
            minx <= x <= maxx
            miny <= y <= maxy
    需 a != 0 且 b != 0; |a|,|b|,|c|, 範圍 ~1e9 內不溢位
*/
ll gcd_ext(ll a, ll b, ll &x, ll &y) {
    if (!b) { x = 1, y = 0; return a; }
    ll x1, y1, g = gcd_ext(b, a % b, x1, y1);
    x = y1, y = x1 - (a / b) * y1;
    return g;
}
bool find_any_solution(ll a, ll b, ll c, ll &x0, ll &y0, ll &g) {
    g = gcd_ext(abs(a), abs(b), x0, y0);
    if (c % g) return false;
    x0 *= c / g, y0 *= c / g;
    if (a < 0) x0 = -x0;
    if (b < 0) y0 = -y0;
    return true;
}
void shift_solution(ll &x, ll &y, ll a, ll b, ll cnt) {
    x += cnt * b;
    y -= cnt * a;
}

ll find_all_solutions(ll a, ll b, ll c, ll minx, ll maxx, ll miny, ll maxy) {
    ll x, y, g;
    if (!find_any_solution(a, b, c, x, y, g))
        return 0;
    a /= g;
    b /= g;

    ll sign_a = a > 0 ? +1 : -1;
    ll sign_b = b > 0 ? +1 : -1;

    shift_solution(x, y, a, b, (minx - x) / b);
    if (x < minx)
        shift_solution(x, y, a, b, sign_b);
    if (x > maxx)
        return 0;
    ll lx1 = x;

    shift_solution(x, y, a, b, (maxx - x) / b);
    if (x > maxx)
        shift_solution(x, y, a, b, -sign_b);
    ll rx1 = x;

    shift_solution(x, y, a, b, -(miny - y) / a);
    if (y < miny)
        shift_solution(x, y, a, b, -sign_a);
    if (y > maxy)
        return 0;
    ll lx2 = x;

    shift_solution(x, y, a, b, -(maxy - y) / a);
    if (y > maxy)
        shift_solution(x, y, a, b, sign_a);
    ll rx2 = x;

    if (lx2 > rx2)
        swap(lx2, rx2);
    ll lx = max(lx1, lx2);
    ll rx = min(rx1, rx2);

    if (lx > rx)
        return 0;
    return (rx - lx) / abs(b) + 1;
}
```

convex-hull.cpp

```cpp
// 注意: 重複點會全部留在輸出裡, 用前先 sort+unique 去重
// 共線邊上的點會保留(非嚴格凸包); |座標|<=1e9 時 cross 不溢位
class Point{
    public:
        ll x,y;
        Point(){}
        Point(ll x,ll y):x(x),y(y){}
        friend ll operator^(const Point a,const Point b){return a.x*b.y-a.y*b.x;}
        friend ll operator*(const Point a,const Point b){return a.x*b.x+a.y*b.y;}
        friend Point operator-(const Point a,const Point b){
            return Point{a.x-b.x,a.y-b.y};
        }
        friend istream& operator>>(istream &in,Point &a){
            in >> a.x >> a.y;
            return in;
        }
        friend ostream& operator<<(ostream &out,Point &a){
            out << a.x << ' ' << a.y << '\n';
            return out;
        }
        friend bool operator<(const Point a,const Point b){return a.x<b.x||(a.x==b.x && a.y<b.y);}
};
ll dis(Point a,Point b){
    return (a.x-b.x)*(a.x-b.x)+(a.y-b.y)*(a.y-b.y);
```

```cpp
}
ll ok(Point a,Point b,Point c){
	ll ang=(b-a)^(c-a); //ang>0 false ang< true
	return ang<0;
}
constexpr ll mxN=2e5+5;
ll st[mxN],id=0;
void solve(istream &cin){
	ll n; cin >> n;
	vc<Point> a(n); for(auto &v:a) cin >> v;
	forn(i,1,n) if(a[i]<a[0]) swap(a[i],a[0]);
	sort(1+all(a),[p=a[0]](Point a,Point b){ll ang=(a-p)^(b-p); return !ang?dis(p,a)<dis(p,b):ang>0;});
	{
		ll j; for(j=n-1;j>=0 && ((a[j]-a[0])^(a[n-1]-a[0]))==0;j--); reverse(j+1+a.bg(),a.ed());
	}
	forn(i,0,n){
		while(id>1 && ok(a[st[id-1]],a[st[id]],a[i])) --id;
		st[++id]=i;
	}
	cout << id << '\n';
	if(id) for(;id;cout << a[st[id--]]);
}
```

cross-product.cpp

```cpp
class Point{
	public:
		ll x,y;
		Point(){}
		Point(ll x,ll y):x(x),y(y){}
		ll f(Point a,Point b){
			return a.x*b.y-b.x*a.y;
		}//croos product
		friend Point operator-(const Point a,const Point b){
			return Point{a.x-b.x,a.y-b.y};
		}
		friend istream& operator>>(istream &in,Point &a){
			in >> a.x >> a.y;
			return in;
		}
};
```

extend-li-chao-tree.cpp

```cpp
// solve()斜率 (y2-y1)/(x2-x1)：垂直線段(x1==x2)會除以零，不整除會截斷
class Line{
	public:
		ll m,c;
		Line(){m=0,c=-inf;} //通用版：空節點 =-inf
		//Line(){m=0,c=-1;} //原題版sentinel：該題「查無值輸出-1」時用這行
		Line(ll m,ll c):m(m),c(c){}
		ll cal(ll x){return m*x+c;}
};
constexpr ll mxN=1e5;
Line t[mxN<<2];
void insert(ll i,ll sl,ll sr,ll l,ll r,Line L){
	if(l>sr || r<sl) return ;
	ll mid=(sl+sr)>>1;
	if(l<=sl && sr<=r){
		if(L.cal(mid)>t[i].cal(mid)) swap(L,t[i]);
		if(sl==sr) return;
		if(L.m>t[i].m) insert(i<<1|1,mid+1,sr,l,r,L);
		else insert(i<<1,sl,mid,l,r,L);
		return ;
	}
	insert(i<<1,sl,mid,l,r,L);
	insert(i<<1|1,mid+1,sr,l,r,L);
}
ll query(ll i,ll sl,ll sr,ll x){
	ll mid=(sl+sr)>>1;
	if(sl==sr) return t[i].cal(x);
	if(x<=mid) return max(t[i].cal(x),query(i<<1,sl,mid,x));
	else return max(t[i].cal(x),query(i<<1|1,mid+1,sr,x));
}
void solve(istream &cin){
	ll n,m,r0=-1; cin >> n >> m;
	forn(i,0,n){
		ll x1,x2,y1,y2; cin >> x1 >> y1 >> x2 >> y2;
		if(!x1) r0=max(r0,y1);
		Line l((y2-y1)/(x2-x1),-x1*(y2-y1)/(x2-x1) + y1);
		insert(1,1,mxN,x1,x2,l);
	}
	cout << r0 << " \n"[m==1];
	forn(i,1,m+1) cout << query(1,1,mxN,i) << " \n"[i==m];
}
```

li-chao-tree.cpp

```cpp
// solve()是特定題：(y2-y1)/m 整數除法，斜率不整除會錯，通用時只抄struct+insert+query
class Line{
	public:
		ll m,c;
		Line(){m=0,c=-inf;} //通用版：空節點=-inf (m=0, cal不會溢位)
		//Line(){m=c=0;}   //原題版：答案保證>=0才可用，否則負答案會被鎖成0
		Line(ll m,ll c):m(m),c(c){}
		ll cal(ll x){return m*x+c;}
};
constexpr ll mxN=1e5;
Line t[mxN<<2];
void insert(ll i,ll sl,ll sr,Line l){
	ll mid=(sl+sr)>>1;
	if(l.cal(mid)>t[i].cal(mid)) swap(l,t[i]);
	if(sl==sr) return;
	if(l.m>t[i].m) insert(i<<1|1,mid+1,sr,l);
	else insert(i<<1,sl,mid,l);
}
ll query(ll i,ll sl,ll sr,ll x){
	ll mid=(sl+sr)>>1;
	if(sl==sr) return t[i].cal(x);
	if(x<=mid) return max(t[i].cal(x),query(i<<1,sl,mid,x));
	else return max(t[i].cal(x),query(i<<1|1,mid+1,sr,x));
}
void solve(istream &cin){
	ll n,m; cin >> n >> m;
	forn(i,0,n){
		ll y1,y2; cin >> y1 >> y2;
		// (y2-y1)/(m-0) = (y-y1)/x
		//x * (y2-y1)/m + y1= y
		Line l((y2-y1)/m,y1);
		insert(1,1,mxN,l);
	}
	forn(i,0,m+1) cout << query(1,1,mxN,i) << " \n"[i==m];
}
```

max-man-dis.cpp

```cpp
constexpr ll mxN=2e5+1;
ll n,d[2][2];
void solve(){
    cin >> n;
    d[0][0]=d[1][0]=-inf;
    d[0][1]=d[1][1]=inf;
    forn(i,0,n){
        ll x,y;
        cin >> x >> y;
        d[0][0]=max(d[0][0],x+y);
        d[0][1]=min(d[0][1],x+y);
        d[1][0]=max(d[1][0],x-y);
        d[1][1]=min(d[1][1],x-y);
        cout << max(d[0][0]-d[0][1],d[1][0]-d[1][1]) << '\n';
    }
}
```

minimum-euc-dis.cpp

```cpp
class Point{
    public:
        ll x,y;
        Point(){}
        Point(ll x,ll y):x(x),y(y){}
        friend ll operator^(const Point a,const Point b){return a.x*b.y-a.y*b.x;}
        friend ll operator*(const Point a,const Point b){return a.x*b.x+a.y*b.y;}
        friend Point operator-(const Point a,const Point b){
            return Point{a.x-b.x,a.y-b.y};
        }
        friend istream& operator>>(istream &in,Point &a){
            in >> a.x >> a.y;
            return in;
        }
        friend bool operator<(const Point a,const Point b){return a.x<b.x||(a.x==b.x && a.y<b.y);}
};
ll euc(Point a,Point b){
    return (a.x-b.x)*(a.x-b.x)+(a.y-b.y)*(a.y-b.y);
}
void solve(istream &cin){
    ll n; cin >> n;
    vc<Point> a(n); for(auto &v:a) cin >> v;
    sort(all(a),[](Point u,Point v){return (u.y<v.y)||(u.y==v.y && u.x<v.x);});
    ll d;
    mls<Point> hp;
    d=euc(a[0],a[1]),hp.emplace(a[0]),hp.emplace(a[1]);
    for(ll i=2,p=0;i<n;i++){
        for(;p<i && (a[p].y-a[i].y)*(a[p].y-a[i].y)>=d;hp.erase(hp.lwb(a[p])),p++); //p<i! 寫p<n重複點(d=0)會清空set後erase(end())炸掉
        auto it=hp.lwb(a[i]);
        auto front=it,back=it;
        forn(j,0,4){
            if(front!=hp.ed()) d=min(d,euc(a[i],*front));
            else break; front++;
        }
        forn(j,0,4){
            if(back!=hp.ed()) d=min(d,euc(a[i],*back));
            {if(back!=hp.bg()) back--; else break;}
        }
        hp.emplace(a[i]);
    }
    cout << d << '\n';
}
```

pick-theorem.cpp

```cpp
void solve(istream &cin){
    ll n; cin >> n;
    vc<Point> a(n);
    for(auto &v:a) cin >> v;
    ll area=0,B=0;
    forn(i,0,n){
        ll j=(i+1)%n;
        area+=a[i]^a[j];
        B+=gcd(abs(a[i].x-a[j].x),abs(a[i].y-a[j].y));
    }
    cout << (abs(area)/2+1-B/2) << ' ' << B << '\n';
    //area=I+B/2-1 I=internal node B=boundary node
}
```

ray-casting.cpp

```cpp
class Point{
    public:
        ll x,y;
        Point(){}
        Point(ll x,ll y):x(x),y(y){}
        friend ll operator^(const Point a,const Point b){return a.x*b.y-a.y*b.x;}
        friend ll operator*(const Point a,const Point b){return a.x*b.x+a.y*b.y;}
        friend Point operator-(const Point a,const Point b){
            return Point{a.x-b.x,a.y-b.y};
        }
        friend istream& operator>>(istream &in,Point &a){
            in >> a.x >> a.y;
            return in;
        }
};
bool on(Point &a,Point &b,Point &c){
        if((b-a)^(c-a)) return false;
        return 0<=((b-a)*(c-a)) && ((b-a)*(c-a))<=((b-a)*(b-a));
}
void query(vc<Point> &a,Point &p){
    ll n=a.size(),ok=0;
    forn(i,0,n){
        ll j=(i+1)%n;
        if(on(a[i],a[j],p)){cout << "BOUNDARY" << '\n'; return;}
        if(p.y>a[i].y!=p.y>a[j].y){
            ll sg=(a[j]-a[i])^(p-a[i]);
            if(sg<0==a[i].y>a[j].y) ok^=1;
        }

    }
    cout << (ok&1?"INSIDE":"OUTSIDE") << '\n';
}
```

動態凸包.cpp

```cpp
struct Point {
    long long x, y;
    bool operator<(const Point& p) const {
        return x < p.x || (x == p.x && y < p.y);
    }
    bool operator==(const Point& p) const { return x == p.x && y == p.y; }
};
```

```cpp
long long cross(const Point& a, const Point& b, const Point& c) {
    // (b-a) x (c-a)
    return (b.x - a.x) * (c.y - a.y) - (b.y - a.y) * (c.x - a.x);
}

// 只支援插入；嚴格凸包（共線點會被移除），每條殼同 x 只留一點
struct DynamicConvexHull {
    set<Point> up, dn; // 上凸殼、下凸殼（皆由左至右）

    void insert(const Point& p) {
        insertHull(up, p, +1);
        insertHull(dn, p, -1);
    }

    // 逆時針凸包頂點（不重複）；1 點/2 點退化也正確
    vector<Point> getConvexHull() const {
        vector<Point> res(dn.begin(), dn.end());
        for (auto it = up.rbegin(); it != up.rend(); ++it)
            if (!(res.size() && (*it == res.back() || *it == res.front())))
                res.push_back(*it);
        return res;
    }

private:
    // side=+1 上殼，-1 下殼；p 在殼上或內側回傳 true
    bool covered(const set<Point>& h, const Point& p, long long side) const {
        if (h.empty() || p.x < h.begin()->x || p.x > h.rbegin()->x) return false;
        auto r = h.lower_bound(Point{p.x, LLONG_MIN}); // 第一個 x>=p.x
        if (r->x == p.x) return side * (r->y - p.y) >= 0;
        return side * cross(*prev(r), *r, p) <= 0;
    }
    void insertHull(set<Point>& h, const Point& p, long long side) {
        if (covered(h, p, side)) return; // 在殼內，不影響凸包
        auto r = h.lower_bound(Point{p.x, LLONG_MIN});
        if (r != h.end() && r->x == p.x) h.erase(r); // 同x被p支配
        auto it = h.insert(p).first;
        while (it != h.begin()) { // 往左刪不再凸的點
            auto b = prev(it);
            if (b == h.begin()) break;
            auto a = prev(b);
            if (side * cross(*a, *b, *it) >= 0) h.erase(b);
            else break;
        }
        while (next(it) != h.end()) { // 往右刪不再凸的點
            auto b = next(it);
            if (next(b) == h.end()) break;
            auto c = next(b);
            if (side * cross(*it, *b, *c) >= 0) h.erase(b);
            else break;
        }
    }
};
```

多邊形重心計算.cpp

```cpp
// 注意：面積為0(退化/全共線)會除以零得NaN；整數座標>~3e7時double精度風險(可改ll累加)
struct Point {
    double x, y;
};

Point polygonCentroid(const vector<Point>& poly) {
    double cx = 0, cy = 0, area = 0;
    int n = poly.size();
    for (int i = 0; i < n; ++i) {
        double cross = poly[i].x * poly[(i + 1) % n].y - poly[(i + 1) % n].x * poly[i].y;
        cx += (poly[i].x + poly[(i + 1) % n].x) * cross;
        cy += (poly[i].y + poly[(i + 1) % n].y) * cross;
        area += cross;
    }
    area /= 2.0;
    cx /= (6.0 * area);
    cy /= (6.0 * area);
    return {cx, cy};
}
```

找三點組成最大圓.cpp

```cpp
// 找三點使外接圓最大：O(n^3) 枚舉（答案的三點通常「不在」凸包上，不能旋轉卡殼）
// 共線三點視為無效（回傳 0）；全部共線輸出 0. n~300 內可跑
#include <bits/stdc++.h>
using namespace std;
typedef long long ll;
typedef pair<ll, ll> P;

ll cross(P a, P b) { return a.first * b.second - a.second * b.first; }
ll dist2(P a, P b) {
    ll dx = a.first - b.first, dy = a.second - b.second;
    return dx * dx + dy * dy;
}
double dist(P a, P b) { return sqrt((double)dist2(a, b)); }

// R = abc / (4S); 2S 用 ll cross 算避免浮點誤判共線
double circumradius(P a, P b, P c) {
    ll S2 = abs(cross({b.first-a.first, b.second-a.second},
                      {c.first-a.first, c.second-a.second})); // 2S
    if (S2 == 0) return 0; // 共線
    return dist(b,c) * dist(a,c) * dist(a,b) / (2.0 * (double)S2);
}

int main() {
    int n;
    cin >> n;
    vector<P> pts(n);
    for (int i = 0; i < n; ++i) cin >> pts[i].first >> pts[i].second;
    double ans = 0;
    for (int i = 0; i < n; ++i)
        for (int j = i + 1; j < n; ++j)
            for (int k = j + 1; k < n; ++k)
                ans = max(ans, circumradius(pts[i], pts[j], pts[k]));
    cout << fixed << setprecision(6) << ans << endl;
}
```

極角排序.cpp

```cpp
struct Point {
    long long x, y;
};

// 以 base 為原點極角排序：從+x軸(角度0)逆時針到2；同角度距離近的在前
// 必須先分上下半平面再比cross，否則超過180度時comparator無效(UB)
// base 本身若在點集中會被排在最前（視為角度 0 距離 0）
void polarSort(vector<Point>& points, Point base) {
    auto half = [&](const Point& p) { // 0:上半平面(含+x軸)，1:下半平面(含-x軸)
        long long dx = p.x - base.x, dy = p.y - base.y;
        return (dy < 0 || (dy == 0 && dx < 0)) ? 1 : 0;
    };
    sort(points.begin(), points.end(), [&](const Point& a, const Point& b) {
        if (half(a) != half(b)) return half(a) < half(b);
```

```cpp
        long long ax = a.x - base.x, ay = a.y - base.y;
        long long bx = b.x - base.x, by = b.y - base.y;
        long long cr = ax * by - ay * bx;
        if (cr != 0) return cr > 0;
        return ax * ax + ay * ay < bx * bx + by * by; // 座標<=1e9不溢位
    });
}
```

5 Data Structure

Array Version of Linked List.cpp

```cpp
// 陣列版雙向串列. 用前必須先 init 串好(0 與 n+1 是 sentinel, 資料放 1..n),
// 否則 cut/insert 解參考 NULL 直接 segfault. cut(p): 把 p 從串列拔出並回傳;
// insert(p,q): 把 q 插到 p 右邊. sentinel 本身不可被 cut.
struct node{
    struct node *nxt, *prv;
    long long v;
}pos[100002];

void init(int n){ // 0 -> 1 -> ... -> n -> n+1
    for(int i=0;i<=n+1;i++){
        pos[i].nxt = (i<=n ? &pos[i+1] : NULL);
        pos[i].prv = (i>=1 ? &pos[i-1] : NULL);
    }
}

inline node* cut(node *p) // cut the node
{
    p->nxt->prv = p->prv;
    p->prv->nxt = p->nxt;
    p->nxt = p->prv = NULL;
    return p;
}

inline node* insert(node *p, node *q) // insert node q to right side of node p
{
    q->nxt = p->nxt;
    p->nxt->prv = q;
    q->prv = p;
    p->nxt = q;
    return q;
}
```

Binary Index Tree.cpp

```cpp
//BIT is always 1 index; built是O(mxN)且用+=累加, 多測要先清t[]並保證a[n+1..]為0
ll n,t[mxN],a[mxN];
void built(){forn(i,1,mxN){ t[i]+=a[i]; if(i+(i&-i)<mxN) t[i+(i&-i)]+=t[i];}} //O(n)
void pad(ll p,ll val){for(;p<mxN;t[p]+=val,p+=p&-p);}
ll query(ll p){ll ans=0; for(;p;ans+=t[p],p-=p&-p); return ans;}
```

Disjoint Set Union.cpp

```cpp
#include <bits/stdc++.h>
using namespace std;

class DisjointSets {
  private:
        vector<int> parents;
        vector<int> sizes;

  public:
        DisjointSets(int size) : parents(size), sizes(size, 1) {
                for (int i = 0; i < size; i++) { parents[i] = i; }
        }

        /** @return the "representative" node in x's component */
        int find(int x) {
                return parents[x] == x ? x : (parents[x] = find(parents[x]));
        }

        /** @return whether the merge changed connectivity */
        bool unite(int x, int y) {
                int x_root = find(x);
                int y_root = find(y);
                if (x_root == y_root) { return false; }

                if (sizes[x_root] < sizes[y_root]) { swap(x_root, y_root); }
                sizes[x_root] += sizes[y_root];
                parents[y_root] = x_root;
                return true;
        }

        /** @return whether x and y are in the same connected component */
        bool connected(int x, int y) { return find(x) == find(y); }
};
```

Segment Tree.cpp

```cpp
//range add range query // sum query
//初始化(1-indexed): a[i]放進 t[n-1+i] (i=1..n) 再呼叫無參數built(); 多測要清t/lz
#define forr(a,b,c) for(ll a=b;a>=c;--a)
#define forn(a,b,c) for(ll a=b;a<c;++a)
constexpr ll mxN=2e5+1;
#pragma GCC target("lzcnt")
ll n,q,t[mxN<<1],lz[mxN]{};
void built(){forr(i,n-1,1)t[i]=t[i<<1]+t[i<<1|1];}
void apply(ll p,ll k,ll val){t[p]+=val*k; if(p<n) lz[p]+=val;}
void push(ll p){for(ll h=63-__builtin_clzll(p);h;h--) if(lz[p>>h]){apply((p>>h)<<1,1<<(h-1),lz[p>>h]),apply((p>>h)<<1|1,1<<(h-1),lz[p>>h]),lz[p>>h]=0;}}
void built(ll p){for(;p>>=1;)if(!lz[p]) t[p]=t[p<<1]+t[p<<1|1];}
void rad(ll l,ll r,ll val){
    ll l0=l+=n-1,r0=r+=n-1,k=1; push(l0),push(r0);
    for(;l<=r;l>>=1,r>>=1,k<<=1){if(l&1) apply(l++,k,val); if(r&1^1) apply(r--,k,val);}
    built(l0),built(r0);
}
ll query(ll l,ll r){ ll ans=0; push(l+=n-1),push(r+=n-1); for(;l<=r;l>>=1,r>>=1){if(l&1) ans+=t[l++]; if(r&1^1) ans+=t[r--];} return ans;} //range query
ll query(ll p){push(p+=n-1); return t[p];} // point query
```

persistence-seg-tree.cpp

```cpp
// 節點數: build用2n-1, 每次單點改約log(n)+1個; mxT要 >= 2n+q*20 (n=q=2e5時約4.2e6)
// 記憶體約 7e6*(8+4+4)  106MB; 多測時 t/lc/rc 殘留值會錯, 需清到 sz 為止
constexpr ll mxN=2e5+1;
constexpr ll mxT=7e6+1;
ll n,q,t[mxT],a[mxN],rt[mxN<<1],rsc,sz; int lc[mxT],rc[mxT];
void built(ll i,ll sl,ll sr){
```

```cpp
        if(sl==sr){t[i]=a[sl]; return;}
        ll mid; mid=(sl+sr)>>1;
        lc[i]=++sz,rc[i]=++sz;
        built(lc[i],sl,mid);
        built(rc[i],mid+1,sr);
        t[i]=t[lc[i]]+t[rc[i]];
}
ll pst(ll pos,ll val,ll i,ll sl,ll sr){
        ll x; x=++sz;
        if(sl==sr){t[x]=val; return x;}
        ll mid; mid=(sl+sr)>>1;
        if(pos<=mid) lc[x]=pst(pos,val,lc[i],sl,mid),rc[x]=rc[i];
        else lc[x]=lc[i],rc[x]=pst(pos,val,rc[i],mid+1,sr);
        t[x]=t[lc[x]]+t[rc[x]];
        return x;
}
ll query(ll l,ll r,ll i,ll sl,ll sr){
        if(l>sr || r<sl) return 0;
        if(l<=sl && sr<=r) return t[i];
        ll mid; mid=(sl+sr)>>1;
        return query(l,r,lc[i],sl,mid)+query(l,r,rc[i],mid+1,sr);
}
void solve(istream &cin){
        cin>>n>>q;
        sz=rsc=0;
        forn(i,1,n+1) cin>>a[i];
        rt[++rsc]=++sz;
        built(rt[1],1,n);
        while(q--){
                ll op; cin>>op;
                if(op==3){
                        ll k; cin>>k;
                        rt[++rsc]=++sz;
                        lc[rt[rsc]]=lc[rt[k]],rc[rt[rsc]]=rc[rt[k]],t[rt[rsc]]=t[rt[k]];
                }
                else if(op==1){
                        ll k,a,x; cin>>k>>a>>x;
                        rt[k]=pst(a,x,rt[k],1,n);
                }
                else{
                        ll k,a,b; cin>>k>>a>>b;
                        cout << query(a,b,rt[k],1,n) << '\n';
                }
        }
}
```

splay-tree.cpp

```cpp
constexpr ll mxN=2e5+3;
ll n,m,rt,t[mxN][2]{},cnt[mxN],val[mxN],f[mxN],lz[mxN]{},vals[mxN];
string s;
ll dir(ll x){return t[f[x]][1]==x;}
void push_down(ll x){
        if(lz[x]){
                swap(t[x][0],t[x][1]);
                if(t[x][0]) lz[t[x][0]]^=1;
                if(t[x][1]) lz[t[x][1]]^=1;
                lz[x]=0;
        }
}
void push(ll x){cnt[x]=cnt[t[x][0]]+1+cnt[t[x][1]],vals[x]=vals[t[x][0]]+val[x]+vals[t[x][1]];}
void rotate(ll x){
        ll y=f[x],z=f[y],d=dir(x);
        t[y][d]=t[x][d^1],t[x][d^1]=y;
        if(t[y][d]) f[t[y][d]]=y;
        if(z) t[z][dir(y)]=x;
        f[y]=x,f[x]=z,push(y),push(x);
}
void splay(ll &z,ll x){
        ll w=f[z];
        for(ll y;(y=f[x])!=w;rotate(x)){
                if(f[y]!=w) rotate(dir(x)==dir(y)?y:x);
        }
        z=x;
}
void build(){
        rt=0;
        forn(i,1,n+3){
                t[i][0]=rt;
                if(rt) f[rt]=i;
                rt=i;
        }
        splay(rt,1);
}
void loc(ll &z,ll k){
        ll x=z;
        for(push_down(x);cnt[t[x][0]]+1!=k;push_down(x)){
                if(k>cnt[t[x][0]])
                        k-=cnt[t[x][0]]+1,x=t[x][1];
                else x=t[x][0];
        }
        splay(z,x);
}
void opt(ll l,ll r){
        loc(rt,l);
        loc(t[rt][1],r-l+2);
        ll x=t[t[rt][1]][0];
        lz[x]^=1;
}
ll query(ll l,ll r){
        loc(rt,l),loc(t[rt][1],r-l+2);
        ll x=t[t[rt][1]][0],y=vals[x];
        return y;
}
void solve(istream &cin){
        cin >> n >> m;
        forn(i,2,n+2) cin >> val[i];
        build();
        forn(i,0,m){
                ll op,l,r; cin >> op >> l >> r;
                if(op==1) opt(l,r);
                else cout << query(l,r) << '\n';
        }
}
```

6 Not that important

EPS.cpp

```cpp
#include <cmath>
const double EPS = 1e-9;

bool eq(double a, double b) {
    return fabs(a - b) < EPS;
}

bool lt(double a, double b) {
    return a < b - EPS;
}

bool le(double a, double b) {
    return a < b + EPS;
}
```

Gauce Elimination.cpp

```cpp
// a[N][N] 是增广矩阵
int gauss()
{
    int c, r;
    for (c = 0, r = 0; c < n; c ++ )
    {
```

```
        int t = r;
        for (int i = r; i < n; i ++ )   // 找到绝对值最大的行
            if (fabs(a[i][c]) > fabs(a[t][c]))
                t = i;

        if (fabs(a[t][c]) < eps) continue;

        for (int i = c; i <= n; i ++ ) swap(a[t][i], a[r][i]);      // 将绝对值最大的行换到最顶端
        for (int i = n; i >= c; i -- ) a[r][i] /= a[r][c];      // 将当前行的首位变成1
        for (int i = r + 1; i < n; i ++ )       // 用当前行将下面所有的列消成0
            if (fabs(a[i][c]) > eps)
                for (int j = n; j >= c; j -- )
                    a[i][j] -= a[r][j] * a[i][c];

        r ++ ;
    }

    if (r < n)
    {
        for (int i = r; i < n; i ++ )
            if (fabs(a[i][n]) > eps)
                return 2; // 无解
        return 1; // 有无穷多组解
    }

    for (int i = n - 1; i >= 0; i -- )
        for (int j = i + 1; j < n; j ++ )
            a[i][n] -= a[i][j] * a[j][n];

    return 0; // 有唯一解
}
```

Joseph.cpp

```
// 回傳 0-indexed 的倖存者位置
ll josephus(ll n, ll k) { // O(n)
  ll res = 0;
  for (ll i = 1; i <= n; ++i) res = (res + k) % i;
  return res;
}
ll josephus2(ll n, ll k) { // O(k log n), k<=n 時用; k>n 落回迴圈版避免深遞迴爆棧
  if (n == 1) return 0;
  if (k == 1) return n - 1;
  if (k > n) {
    ll res = 0;
    for (ll i = 2; i <= n; ++i) res = (res + k) % i;
    return res;
  }
  ll res = josephus2(n - n / k, k);
  res -= n % k;
  if (res < 0)
    res += n;  // mod n
  else
    res += res / (k - 1);  // 还原位置
  return res;
}
```

7 New

Convex_hull_trick.cpp

```
// 特定題解法: 斜率=(y2-y1)/m 是整數除法(負數向零截斷), 查詢x需單調遞增
class line{
        public:
                ll m,c;
                line(){}
                line(ll m,ll c):m(m),c(c){}
                friend bool operator<(const line &a,const line &b){
                        return a.m<b.m || (a.m==b.m && a.c>b.c);
                }
                ll cal(ll x){return m*x+c;}
};
constexpr ll mxN=1e5+1;
ll n,m,id=0,st[mxN];
vc<line> ln;
bool check(ll i,ll j,ll k){ //__int128防溢位(c差*m差可到1e18以上)
        return (__int128)(ln[i].c-ln[k].c)*(ln[j].m-ln[i].m)<(__int128)(ln[i].c-ln[j].c)*(ln[k].m-ln[i].m);
}
void solve(istream &cin){
        cin >> n >> m;
        forn(i,0,n){
                ll y1,y2; cin >> y1 >> y2;
                ln.emp(line{(y2-y1)/m,y1});
        }
        sort(all(ln)),ln.resize(unique(all(ln),[](auto a,auto b){return a.m==b.m;})-ln.bg());
        forn(i,0,ln.size()){
                while(id>1 && check(st[id-1],st[id],i)) id--;
                st[++id]=i;
        }
        ll ptr=1;
        forn(i,0,m+1){
                while(ptr+1<=id && ln[st[ptr+1]].cal(i)>ln[st[ptr]].cal(i)) ptr++;
                cout << ln[st[ptr]].cal(i) << " \n"[i==m];
        }
}
```

Ray_casting.cpp

```
class Point{
        public:
                ll x,y;
                Point(){}
                Point(ll x,ll y):x(x),y(y){}
                friend Point operator-(const Point &a,const Point &b){return {a.x-b.x,a.y-b.y};}
                friend ll operator*(const Point &a,const Point &b){return a.x*b.x+a.y*b.y;}
                friend ll operator^(const Point &a,const Point &b){return a.x*b.y-a.y*b.x;}
};
ll n,m;
vc<Point> p;
bool on(Point &a,Point &b,Point &c){
        if((b-a)^(c-a)) return false;
        return 0<=((b-a)*(c-a)) && ((b-a)*(c-a))<=((b-a)*(b-a));
}
void query(Point &x){
        ll ok=0;
        forn(i,0,n){
                if(on(p[i],p[(i+1)%n],x)){ cout << "BOUNDARY" << '\n';return;}
                Point &a=p[i],&b=p[(i+1)%n];
                if(a.y>x.y != b.y>x.y){
                        ll sg=(x-a)^(b-a);
                        if(b.y>a.y==(sg<0)) ok^=1;
                }
        }
        cout << (ok&1?"INSIDE":"OUTSIDE") << '\n';
}
```

block_cut_tree.cpp

```cpp
constexpr ll mxN=1e5+1;
constexpr ll mxB=20;
ll n,m,q,st[mxN],id=0,mark[mxN]{},dfn[mxN]{},low[mxN],tim=0,nid[mxN],pa[mxN<<1][mxB],d[mxN<<1];
vc<ll> adj[mxN];
vc<vc<ll>> cmp,t;
void tarjan(ll u=1,ll p=0){
	dfn[u]=low[u]=++tim,st[++id]=u;
	for(auto&v:adj[u]){
		if(v==p) continue;
		if(dfn[v]) low[u]=min(low[u],dfn[v]);
		else{
			tarjan(v,u);
			low[u]=min(low[u],low[v]);
			if(low[v]>=dfn[u]){
				mark[u]=(dfn[u]>1 || dfn[v]>2);
				cmp.emp(vc<ll>{u});
				while(cmp.back().back()!=v) cmp.back().emp(st[id--]);
			}
		}
	}
}
void dfs(ll u=1,ll p=0,ll lv=1){
	d[u]=lv,pa[u][0]=p;
	for(auto&v:t[u]){
		if(v!=p) dfs(v,u,lv+1);
	}
}
void built(){
	ll x=0;
	t.emp(vc<ll>{});
	forn(i,1,n+1) if(mark[i]) nid[i]=++x,t.emp(vc<ll>{});
	for(auto&a:cmp){
		ll y=++x;
		t.emp(vc<ll>{});
		for(auto&u:a){
			if(mark[u]) t[nid[u]].emp(y),t[y].emp(nid[u]);
			else nid[u]=y;
		}
	}
	dfs();
	forn(j,1,mxB) forn(i,1,t.size()) pa[i][j]=pa[pa[i][j-1]][j-1];
}
ll lca(ll a,ll b){
	if(d[a]<d[b]) swap(a,b);
	forr(i,mxB-1,0) if(d[a]-(1<<i)>=d[b]) a=pa[a][i];
	if(a==b) return a;
	forr(i,mxB-1,0) if(pa[a][i]!=pa[b][i]) a=pa[a][i],b=pa[b][i];
	return pa[a][0];
}
ll query(ll a,ll b,ll c){
	ll lca1=lca(a,b),lca2=lca(a,c),lca3=lca(b,c);
	return (lca1==c || (lca2==c && lca3==lca1) || (lca3==c && lca2==lca1));
}
void solve(){
	cin >> n >> m >> q;
	forn(i,0,m){
		ll u,v; cin >> u >> v;
		adj[u].emp(v),adj[v].emp(u);
	}
	tarjan();
	built();
	while(q--){
		ll a,b,c; cin >> a >> b >> c;
		cout << ((mark[c] && query(nid[a],nid[b],nid[c])) || a==c || b==c?"NO":"YES") << '\n';
	}
}
```

cross_dot_product.cpp

```cpp
//A=(x1,y1),B=(x2,y2)
//A cross B = ABsin x = x1*y2-y1*x2
//A dot B =AB cos x = x1*x2+y1*y2
```

dec_string_to_bitset.cpp

```cpp
// 10進位string(只含0~9, 長度<=500) 轉 bitset 二進位, 支援先減一個小常數
// decSub(s,x): 回傳 s-x (x>=0, 需保證 s>=x), O(len)
// decToBits(s): 10進string -> bitset, b[0]為LSB, O(len^2)
// bitsToDec(b): bitset -> 10 進 string, 驗證/輸出用
// 500 位 10 進  1661 bits, mxB=2048 夠用
constexpr ll mxB=2048;
string decSub(string s,ll x){
	for(ll i=s.size()-1;i>=0&&x;i--){
		ll d=(s[i]-'0')-x%10; x/=10;
		if(d<0){d+=10; x++;}
		s[i]=d+'0';
	}
	ll p=0; while(p+1<(ll)s.size()&&s[p]=='0') p++;
	return s.substr(p);
}
bitset<mxB> decToBits(string s){
	bitset<mxB> b;
	for(ll i=0;s!="0";i++){
		b[i]=(s.back()-'0')&1;
		ll r=0;
		for(char &c:s){ll d=r*10+(c-'0'); c='0'+d/2; r=d&1;}
		ll p=0; while(p+1<(ll)s.size()&&s[p]=='0') p++;
		s=s.substr(p);
	}
	return b;
}
string bitsToDec(bitset<mxB> b){
	string s="0";
	ll hi=mxB-1; while(hi>0&&!b[hi]) hi--;
	forr(i,hi,0){
		ll c=b[i];
		forr(j,(ll)s.size()-1,0){ll d=(s[j]-'0')*2+c; s[j]='0'+d%10; c=d/10;}
		if(c) s=char('0'+c)+s;
	}
	return s;
}
void solve(istream &cin){
	string s; ll k;
	cin>>s>>k; // 10 進 string 與 要減的常數
	bitset<mxB> b=decToBits(decSub(s,k));
	ll hi=mxB-1; while(hi>0&&!b[hi]) hi--;
	forr(i,hi,0) cout<<b[i]; // MSB->LSB 印出 2 進位
	cout<<'\n';
}
```

dominator_tree.cpp

```cpp
constexpr ll mxN=1e5+1;
ll n,m,tim=0,id[mxN],sd[mxN],pa[mxN],f[mxN],lb[mxN],et[mxN]{},ret[mxN];
vc<ll> adj[mxN],radj[mxN],t[mxN],st;
void dfs(ll u=1){
	et[u]=++tim,ret[tim]=u;
	id[tim]=sd[tim]=lb[tim]=pa[tim]=tim;
	for(auto&v:adj[u]){
```

```cpp
            if(!et[v]) dfs(v),f[et[v]]=et[u];
            radj[et[v]].emp(et[u]);
        }
}
ll find(ll u,ll x=0){
        if(u==pa[u]) return x?-1:u;
        ll v=find(pa[u],x+1);
        if(v<0) return u;
        if(sd[lb[u]]>sd[lb[pa[u]]]) lb[u]=lb[pa[u]];
        pa[u]=v;
        return x?v:lb[u];
}
void built(){
        vc<ll> arr[n+1];
        forr(i,n,1){
            for(auto&v:radj[i]) sd[i]=min(sd[i],sd[find(v)]);
            if(i>1) arr[sd[i]].emp(i);
            for(auto&v:arr[i]){
                ll x=find(v);
                if(sd[x]==sd[v]) id[v]=sd[v];
                else id[v]=x;
            }
            if(i>1) pa[i]=f[i];
        }
        forn(i,2,n+1){
            if(id[i]!=sd[i]) id[i]=id[id[i]];
            t[ret[i]].emp(ret[id[i]]),t[ret[id[i]]].emp(ret[i]);
        }
}
void dfs_ans(ll u=1,ll p=0){
        st.emp(u);
        if(u==n){
            sort(all(st));
            cout << st.size() << '\n';
            for(auto&v:st) cout << v << ' '; cout << '\n';
            exit(0);
        }
        for(auto&v:t[u]){
            if(v==p) continue;
            dfs_ans(v,u);
        }
        st.pop_back();
}
void solve(istream &cin){
        cin>>n>>m;
        forn(i,1,m+1){
            ll u,v; cin>>u>>v;
            adj[u].emp(v);
        }
        dfs();
        built();
        dfs_ans();
}
```

extra.cpp

```cpp
//__builtin_popcountll
//__builtin_parityll
//__builtin_clzll
//__builtin_ctzll
// mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
// const ll M = 991831889;
// const ll C = uniform_int_distribution<ll>(0.1 * M, 0.9 * M)(rng);

```

fft.cpp

```cpp
// 兩模數NTT+CRT: 結果係數必須 < 998244353*1004535809 (~1.0027e18), 否則需三模
// sz上限2^21 (n+m+2 <= 2097152); solve()讀n+1,m+1個係數(n,m是次數)
class FFT{
        public:
                ll mod,root,root_inv,maX,M,M_inv;
                FFT(){}
                FFT(ll a,ll b,ll c,ll d,ll e,ll f):mod(a),root(b),root_inv(c),maX(d),M(e),M_inv(f){}
                ll fsp(ll a,ll m){
                        ll ans=1;
                        while(m){
                                if(m&1) ans=(ans*a)%mod;
                                a=(a*a)%mod,m>>=1;
                        }
                        return ans;
                }
                void fft(ll sz,vc<ll> &arr,ll inv){
                        for(ll i=1,j=0;i<sz;++i){
                                ll bit=sz>>1; for(;j&bit;bit>>=1) j^=bit; j^=bit;
                                if(i<j) swap(arr[i],arr[j]);
                        }
                        for(ll l=2;l<=sz;l<<=1){
                                ll wlen=inv?root_inv:root;
                                for(ll i=l;i<maX;i<<=1) wlen=(wlen*wlen)%mod;
                                for(ll i=0;i<sz;i+=l){
                                        ll w=1;
                                        for(ll j=0;j<l/2;++j){
                                                ll x=arr[i+j],y=(w*arr[i+j+l/2])%mod;
                                                arr[i+j]=(x+y<mod?x+y:x+y-mod);
                                                arr[i+j+l/2]=(x-y>=0?x-y:x-y+mod);
                                                w=(w*wlen)%mod;
                                        }
                                }
                        }
                        if(inv){
                                ll n_1=fsp(sz,mod-2);
                                forn(i,0,sz){
                                        arr[i]=(arr[i]*n_1)%mod;
                                }
                        }
                }
};
constexpr ll MM=1002772198720536577;
//
//g=3
//998244353=119*2^23 + 1
//g^c=15311432
//inv=469870224
//M1=1004535809
//M1^-1=332747959
//
//1004535809=479*2^21 + 1
//g^c=702606812
```

```cpp
//inv=700146880
//M2=998244353
//M2^-1=669690699
ll mul(ll a,ll b){
	__int128 aa=a,bb=b,cc;
	cc=(aa*bb)%MM;
	return (ll)cc;
}
void solve(){
	FFT fft1(998244353,15311432,469870224,1<<23,1004535809,332747959);
	FFT fft2(1004535809,702606812,700146880,1<<21,998244353,669690699);
	//ll x=(fft1.M*fft1.M_inv)%fft1.mod;
	//ll y=(fft2.M*fft2.M_inv)%fft2.mod;
	//err(x,y);
	ll n,m;
	cin >> n >> m;
	vc<ll> a(n+1),b(m+1);
	for(auto &v:a) cin >> v;
	for(auto &v:b) cin >> v;
	ll sz=1;
	for(;sz<a.size()+b.size();sz<<=1);
	a.resize(sz),b.resize(sz);
	vc<ll> A=a,B=b;
	fft1.fft(sz,a,0);
	fft1.fft(sz,b,0);
	forn(i,0,sz) a[i]=(a[i]*b[i])%fft1.mod;
	fft1.fft(sz,a,1);

	fft2.fft(sz,A,0);
	fft2.fft(sz,B,0);
	forn(i,0,sz) A[i]=(A[i]*B[i])%fft2.mod;
	fft2.fft(sz,A,1);

	forn(i,0,n+m+1) cout << (mul(a[i],fft1.M*fft1.M_inv) + mul(A[i],fft2.M*fft2.M_inv))%MM << " \n"[i==n+m];
}
```

fwt.cpp

```cpp
constexpr ll mxB=21;
ll a[1<<mxB]{};
void fwt(ll *arr,ll type){
	for(ll k=2;k<=(1<<mxB);k<<=1){ //必須<=, 否則最高層蝶形沒做, 值>=2^(mxB-1)會錯
		for(ll i=0,m=k>>1;i+m<(1<<mxB);i+=k){
			forn(j,0,m){
				ll x,y; x=arr[i+j],y=arr[i+j+m];
				arr[i+j]=x+y,arr[i+j+m]=x-y;
				if(type) arr[i+j]/=2,arr[i+j+m]/=2;
			}
		}
	}
}
void solve(istream &cin){
	ll n,pfx,mx;
	cin>>n;
	mx=pfx=0;
	a[0]=1;
	forn(i,0,n){
		ll x; cin>>x;
		pfx^=x,a[pfx]++,mx=max(mx,a[pfx]);
	}
	fwt(a,0);
	forn(i,0,1<<mxB) a[i]=a[i]*a[i];
	fwt(a,1);
	ll res; res=0;
	// forn(i,0,17) cout << a[i] << " \n"[i==16];
	forn(i,0,1<<mxB) res+=(!i?mx>1:a[i]>0);
	cout << res << '\n';
	forn(i,0,1<<mxB) if(!i?mx>1:a[i]) cout << i << ' '; cout << '\n';
}
```

kmp-automaton.cpp

```cpp
void kmp(){
	forn(i,1,m){
		ll j; j=k[i-1];
		while(j>0 && s[j]!=s[i]) j=k[j-1];
		if(s[j]==s[i]) k[i]=j+1;
	}
}
void solve(istream &cin){
	cin>>n>>s;
	m=s.length();
	kmp();
	//forn(i,0,m){err(i,k[i]);}
	forn(i,0,m+1){
		forn(j,0,26){
			if(i>0 && (s[i]-'A')!=j) nex[i][j]=nex[k[i-1]][j];
			else nex[i][j]=i+(s[i]-'A'==j);
		}
	}
	// forn(i,0,m+1){forn(j,0,26){err(i,j,nex[i][j]);}}
	dp[0][0][0]=1;
	forn(i,0,n){
		forn(f,0,2){
			forn(j,0,m+1){
				forn(c,0,26){
					// err(i,j,c,nex[j][c],dp[i][j]);
					ll nf; nf=f|(nex[j][c]==m);
					dp[i+1][nex[j][c]][nf]=(dp[i+1][nex[j][c]][nf]+dp[i][j][f])%mdl1;
				}
			}
		}
	}
	ll res; res=0;
	forn(i,0,m+1) res=(res+dp[n][i][1])%mdl1;
	cout << res << '\n';
}
```

li_chao_tree.cpp

```cpp
class line {
	public:
	ll m,c;
	line(){m=0,c=-inf-1;}
	line(ll m,ll c):m(m),c(c){}
	ll cal(ll x){return m*x+c;}
};
constexpr ll mxN=1e5+5;
ll n;
line t[mxN<<2];
void insert(line x,ll l,ll r,ll i,ll sl,ll sr){
	if(sr<l || r<sl || l>r) return;
	ll mid=(sl+sr)>>1;
	if(l<=sl && sr<=r){ //if extended otherwise remove this condtion
		bool lf=t[i].cal(sl-1)<x.cal(sl-1),mf=t[i].cal(mid-1)<x.cal(mid-1),rf=t[i].cal(sr-1)<x.cal(sr-1);
		if(mf) swap(x,t[i]);
		if(sl==sr) return;
```

```
            if(lf^mf) insert(x,l,r,i<<1,sl,mid);
            if(rf^mf) insert(x,l,r,i<<1|1,mid+1,sr);
            return;
        }
        insert(x,l,r,i<<1,sl,mid);
        insert(x,l,r,i<<1|1,mid+1,sr);
}
ll query(ll x,ll i,ll sl,ll sr){
        ll mid=(sl+sr)>>1;
        if(sl==sr) return t[i].cal(x-1);
        if(x<=mid) return max(t[i].cal(x-1),query(x,i<<1,sl,mid));
        else return max(t[i].cal(x-1),query(x,i<<1|1,mid+1,sr));
}
void solve(){
        cin >> n;
        forn(i,0,n){
                ll op;
                cin >> op;
                if(op==1){
                        ll a,b,l,r;
                        cin >> a >> b >> l >> r;
                        line x{a,b};
                        insert(x,l+1,r+1,1,1,mxN-4);
                }
                else{
                        ll x;
                        cin >> x;
                        x=query(x+1,1,1,mxN-4);
                        if(x>-inf-1) cout << x << '\n';
                        else cout << "NO" << '\n';
                }
        }
}
```

manacher.cpp

```
ll dp[mxN][2]{},n,res[mxN];
void solve(istream &cin){
        cin >> s,n=s.length();
        ll l1,l2,r1=-1,r2=-1;
        forn(i,0,n) res[i]=1;
        forn(i,0,n){
                if(r1>i) dp[i][1]=min(r1-i,dp[l1+r1-i][1]);
                if(r2>i) dp[i][0]=min(r2-i,dp[l2+r2-i-1][0]);
                while(i+dp[i][1]<n && i-dp[i][1]>=0 && s[i+dp[i][1]]==s[i-dp[i][1]]) res[i+dp[i][1]]=max(res[i+dp[i][1]],2*dp[i][1]+1),dp[i][1]++;
                while(i+dp[i][0]+1<n && i-dp[i][0]>=0 && s[i+dp[i][0]+1]==s[i-dp[i][0]]) res[i+dp[i][0]+1]=max(res[i+dp[i][0]+1],2*(dp[i][0]+1)),dp[i][0]++;
                if(i+dp[i][1]>r1) l1=i-dp[i][1],r1=i+dp[i][1];
                if(i+dp[i][0]>r2) l2=i-dp[i][0]+1,r2=i+dp[i][0];
        }
        forn(i,0,n) cout << res[i] << " \n"[i==n-1];
}
```

shoelace.cpp

```
//consecutive point p[i] (0<=i<n)
//注意：輸出是「2倍面積」(鞋帶公式沒除2); 2A = sum(p[i]^p[i+1])
//| 座標 |<=1e9 時 2A<=8e18 勉強不溢位，更大要 __int128

void solve(istream &cin){
        ll n,res=0;
        cin>>n;
        forn(i,1,n+1){
                cin>>a[i];
                if(i>1) res+=a[i-1]^a[i];//cross product
                if(i==n) res+=a[n]^a[1];
        }
        cout << abs(res) << '\n';
}
```

suffix_automaton.cpp

```
// 靜態陣列約45MB(64MB限制小心); 多次呼叫sam()時sz要歸零(solve有做)
constexpr ll mxN=1e5+1;
class state{
    public:
        ll len,link,next[26];

        state(){}
};
ll n,sz;
state t[mxN<<1];
void sam(const string &s){
    memset(t,0,sizeof(t));
    t[0].link=-1;
    ll last,cur; last=0;
    for(auto&v:s){
        cur=++sz;
        t[cur].len=t[last].len+1;
        ll p; p=last;
        while(p!=-1 && !t[p].next[v-'a']) t[p].next[v-'a']=cur,p=t[p].link;
        if(p==-1) t[cur].link=0;
        else{
            ll q; q=t[p].next[v-'a'];
            if(t[p].len+1==t[q].len) t[cur].link=q;
            else{
                ll clone; clone=++sz;
                t[clone].len=t[p].len+1,t[clone].link=t[q].link;
                memcpy(t[clone].next,t[q].next,sizeof(t[q].next));
                while(p!=-1 && t[p].next[v-'a']==q) t[p].next[v-'a']=clone,p=t[p].link;
                t[q].link=t[cur].link=clone;
            }
        }
        last=cur;
    }
}
void solve(istream &cin){
    string s;
    cin>>s;
    n=s.length(),sz=0;
    sam(s);
    vc<ll> res(n+2,0);
    forn(i,1,sz+1) res[t[t[i].link].len+1]++,res[t[i].len+1]--;
    partial_sum(all(res),res.bg());
    forn(i,1,n+1) cout << res[i] << " \n"[i==n];
}
```

suffixarray_and_lcp.cpp

```
constexpr ll mxN=1e5+5;
ll sa[mxN],nsa[mxN],c[mxN],nc[mxN],cnt[mxN]{},lcp[mxN];
void sfa(string s){
        s+="$";
        ll n=s.length();
        memset(cnt,0,sizeof(cnt)); //cnt有殘留，不清第二次呼叫必錯
        forn(i,0,n) cnt[s[i]]++; forn(i,1,256) cnt[i]+=cnt[i-1];
        forr(i,n-1,0) sa[--cnt[s[i]]]=i; c[sa[0]]=0; forn(i,1,n) c[sa[i]]=c[sa[i-1]]+(s[sa[i]]!=s[sa[i-1]]);
```

```cpp
        for(ll h=0;(1<<h)<n;h++){
                forn(i,0,n) nsa[i]=((sa[i]-(1<<h))%n+n)%n;
                fill(cnt,cnt+c[sa[n-1]]+1,0);
                forn(i,0,n) cnt[c[nsa[i]]]++; forn(i,1,c[sa[n-1]]+1) cnt[i]+=cnt[i-1];
                forr(i,n-1,0) sa[--cnt[c[nsa[i]]]]=nsa[i];
                nc[sa[0]]=0;
                forn(i,1,n){
                        pll pre={c[sa[i-1]],c[(sa[i-1]+(1<<h))%n]},cur={c[sa[i]],c[(sa[i]+(1<<h))%n]};
                        nc[sa[i]]=nc[sa[i-1]]+(cur!=pre);
                }
                swap(c,nc);
        }
}
void Lcp(const string &s){
        ll n=s.length(),k=0;
        forn(i,0,n){
                if(c[i]>n-1){k=0; continue;}
                ll j=sa[c[i]+1];
                for(;i+k<n && j+k<n && s[i+k]==s[j+k];k++);
                lcp[c[i]]=k;
                if(k) k--;
        }
}
```

xor_basis.cpp

```cpp
// 值需 < 2^31; 值域到 1e18 時 mxB 改 60、迴圈上限改 59
constexpr ll mxN=2e5+1;
constexpr ll mxB=31;
ll n,a[mxN],b[mxB]{};
void built(){
        forn(i,1,n+1){
                ll x; x=a[i];
                forr(j,30,0){
                        if((x>>j)&1){
                                if(!b[j]){b[j]=x; break;}
                                else x^=b[j];
                        }
                }
        }
}
void solve(istream &cin){
        cin>>n;
        forn(i,1,n+1) cin>>a[i];
        built();
        ll res; res=0;
        forr(i,30,0){
                if(res>>i&1^1) res^=b[i];
        }
        cout << res << '\n';
}
```

zfunction_and_kmp.cpp

```cpp
constexpr ll mxN=1e6+1;
vc<ll> zf(const string &s){
        ll n,l,r; n=s.length(),l=r=-1;
        vc<ll> z(n,0);
        forn(i,1,n){
                if(r>i) z[i]=min(z[i-l],r-i);
                while(i+z[i]<n && s[i+z[i]]==s[z[i]]) z[i]++;
                if(i+z[i]>r) r=i+z[i],l=i;
        }
        return z;
}
vc<ll> kmp(const string &s){
        ll n; n=s.length();
        vc<ll> k(n,0);
        forn(i,1,n){
                ll j; j=k[i-1];
                while(j>0 && s[i]!=s[j]) j=k[j-1];
                if(s[i]==s[j]) k[i]=j+1;
        }
        return k;
}
```